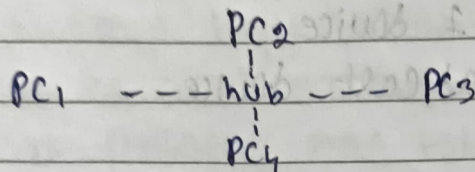


Network device:-

1) Explain the behavior of hub?



* It is used to connect multiple network device.

* Ex:- we have 4 PC like, PC₁, PC₂, PC₃, PC₄

* Now, we send data from PC₁ to PC₄, when the data send to PC₁ to PC₄.

* The hub will broadcast the data to all the PC Port.

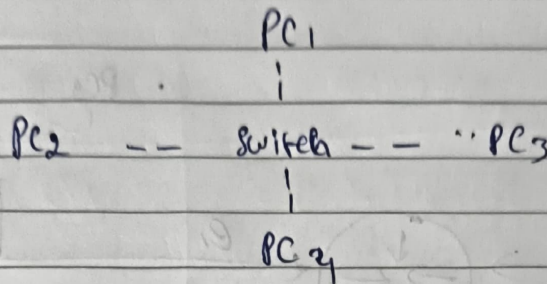
* But the data only open at PC₄ other PC₃ & PC₂ will receive the data but they didn't open the data they can't & it will drop the data.

* It is called an half duplex.

* It is a layer 1 device. when the data receive - It repeats the data through out all the other ports without checking the destination Mac Address.

* The main thing is once the data will shared to hub it will broadcast & collision, because of data loss & heavy traffic.

2) Explain the behavior of switch?



* It is used to connect multiple network device

* It contains mac ~~address~~ Table.

* It will ~~not~~ send data in broadcast & unicast.

* If the data coming ~~for~~ to switch.

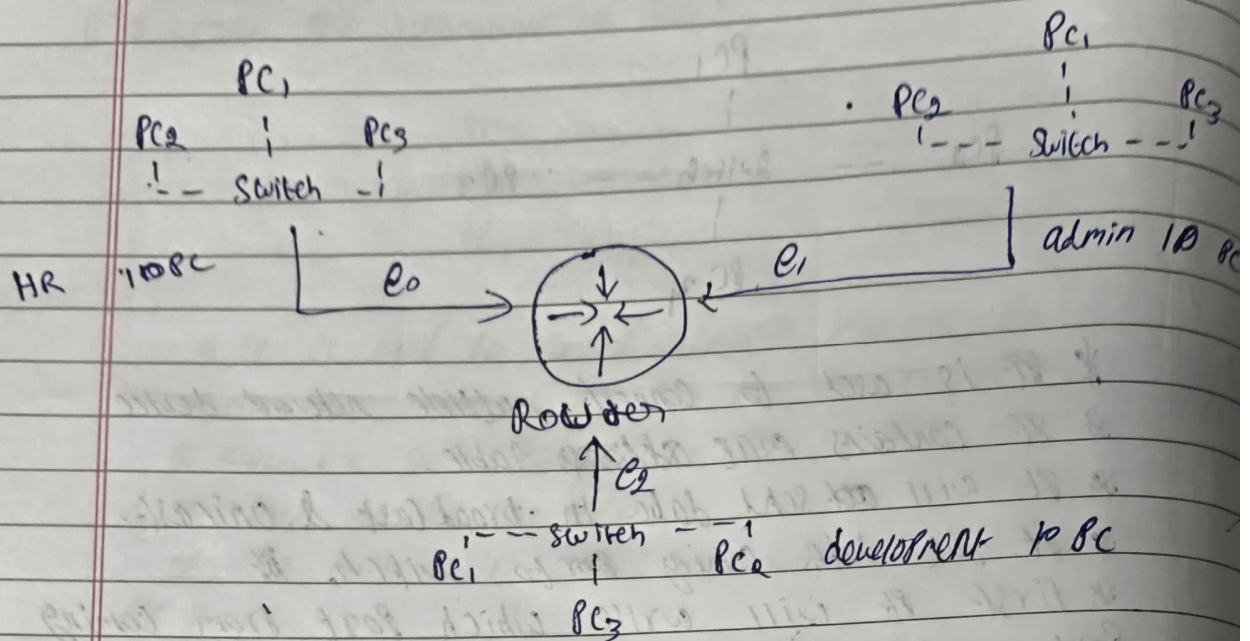
* First it will write which port from coming port number & source mac it is called as address learning.

* When ever the data will come to switch it will check ~~source~~ destination mac in mac table it will detect the destination mac it will send data to that particular port directly it is called unicast.

* open wise destination ~~data~~ doesn't there means, the switch will send all other ports, it is called broadcast.

* It is a intelligent forwarding decision.

5) Explain the Behavior of Router?



It is used to break the broadcast domain.

It contains route table.

It is a unicast device.

We have 3 departments that each connect in switch and that the switches are connected in Router interface.

Here we share data from one department PC to another department PC.

It contains route table it contains IP address, prefix, interface.

When ever we send the data to Router it will check the destination IP address in system route table and if the destination will detect the data send to destination PC or other wise the data drops.

It is used to reduce the traffic while sending data.